

Network Traffic Investigation Timeline

Milestone 2: AI-Assisted Threat Analysis

Dataset: 2021-08-16-formbook.pcap

Analysis method: PCAP conversion → CSV segmentation → AI-assisted analysis → Validation

Section 1: Timeline Overview

This timeline documents the chronological investigation process performed during the AI-assisted analysis of the PCAP network capture. It does not represent confirmed malicious activity; instead, it records the key investigation events, observed network behaviours, AI-assisted findings, and validation steps carried out during the analysis.

The PCAP data was divided into three segments, analyzed individually, and then combined into a final assessment. The table in Section 2 lists each step in the order it occurred, along with the evidence gathered at that step and its security interpretation.

Section 2: Investigation Timeline Table

Step	Investigation Event	Evidence / Observation	Security Interpretation
1	PCAP file obtained for analysis	Source file: 2021-08-16-formbook.pcap	Network capture prepared for investigation
2	PCAP converted into CSV format using tshark	Packet fields extracted including IP addresses, ports, protocols, timestamps and frame information	Created structured dataset suitable for AI analysis
3	CSV validation performed	Converted dataset checked for rows, columns and data consistency	Confirmed data was ready for analysis
4	Dataset divided into three analysis segments	traffic_segment_1.csv, traffic_segment_2.csv, traffic_segment_3.csv	Enabled structured AI-assisted investigation
5	Segment 1 AI analysis performed	Internal host 10.10.0.33 identified as major communicating endpoint	Initial investigation focus identified
6	Segment 1 external communication reviewed	High-volume communication with 18.184.26.60 observed	External relationship required validation
7	Segment 2 AI analysis performed	Repeated communication between 10.10.0.33 and 18.184.26.60 observed	Possible automated application behaviour investigated
8	Segment 3 AI analysis performed	Multiple external services and DNS/HTTP/HTTPS traffic identified	Additional communication patterns analysed

Step	Investigation Event	Evidence / Observation	Security Interpretation
9	Combined AI findings created	Findings from all three segments correlated	Overall traffic behaviour assessed
10	Final validation performed	AI findings reviewed against available dataset evidence	Confirmed observations separated from assumptions

Steps 1 through 4 cover data preparation: obtaining the capture, converting it to a structured CSV format with tshark, validating the resulting dataset, and splitting it into three segments for individual review. Steps 5 through 8 cover the AI-assisted analysis performed on each segment, which is where the internal host 10.10.0.33 and its communication with 18.184.26.60 first became the focus of the investigation. Steps 9 and 10 cover the combination of the three segments into a single assessment and the final validation of the AI-generated findings against the underlying evidence.

Section 3: Key Investigation Findings

Investigation Area	Result
Primary internal host	10.10.0.33
Main external communication	18.184.26.60
Common protocols	TCP, HTTP, HTTPS/TLS, DNS
Beaconing assessment	No confirmed beaconing behaviour
Data exfiltration	No confirmed evidence
Malware confirmation	Not possible using metadata only
Recommended next steps	DNS analysis, payload inspection, endpoint investigation

These findings summarize the outcome of the ten-step investigation process. 10.10.0.33 and its relationship with 18.184.26.60 remain the primary items requiring follow-up, and none of the metadata reviewed across the three segments was sufficient on its own to confirm either beaconing behaviour or malware activity.

Section 4: Visual Timeline Summary

The diagram below summarizes the six-stage investigation pipeline at a high level, from initial PCAP collection through to final assessment. It condenses the ten-step timeline in Section 2 into the major phases of the investigation: PCAP collection, PCAP conversion, traffic segmentation, AI-assisted analysis, threat validation, and final assessment.

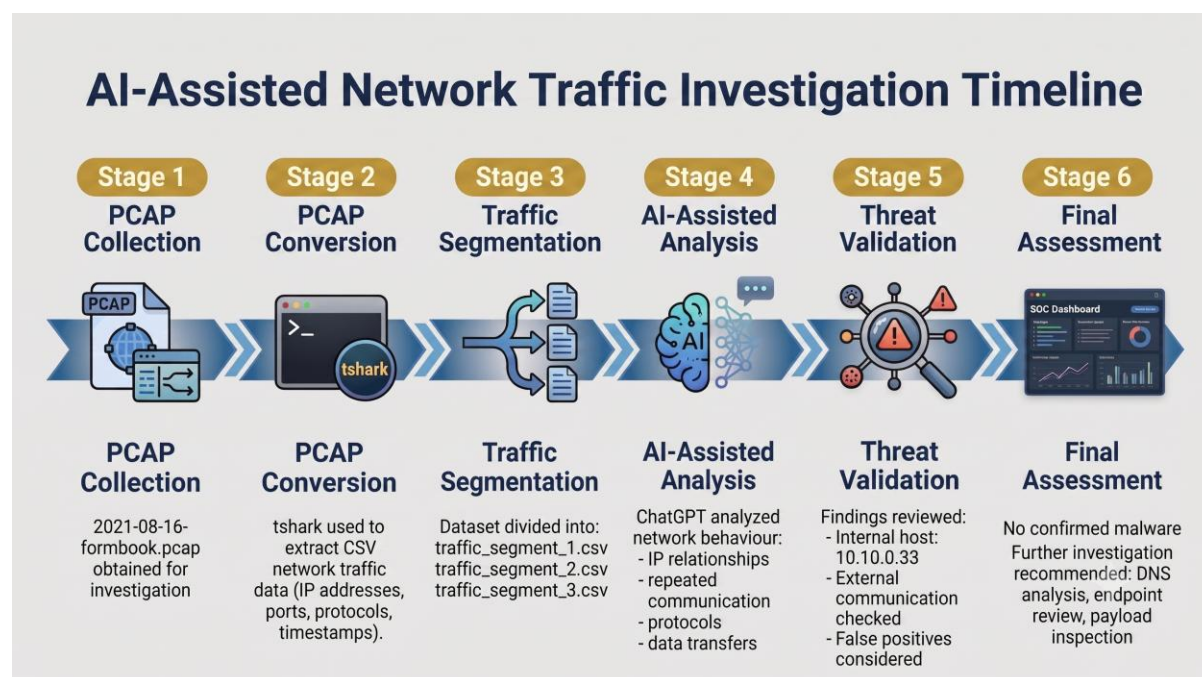


Figure 1: AI-Assisted Network Traffic Investigation Timeline (six-stage summary)

Section 5: Timeline Conclusion

The investigation timeline shows the complete workflow followed during the AI-assisted network traffic analysis process. The analysis identified 10.10.0.33 as the primary internal host communicating with external services, particularly 18.184.26.60.

Repeated communication patterns and high-volume traffic were identified between these two hosts, but the available packet metadata was insufficient to confirm malicious activity. Further investigation using packet payload analysis, DNS inspection, threat intelligence, and endpoint investigation would be required for a final determination.